

LEXICAL ANALYZER PROJECT REPORT

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Project Title: Lexical Analyzer

1. Introduction

A Lexical Analyzer is the first phase of a compiler. It reads source code, groups characters into tokens, and classifies them as keywords, identifiers, operators, constants, special symbols, and string literals.

2. Objective

To develop a C program that scans a C source file and identifies different lexical tokens.

3. Features

- Reads a C source file.
- Identifies keywords.
- Detects identifiers.
- Recognizes operators.
- Identifies special symbols.
- Detects numeric constants.
- Detects string literals.

4. Technologies Used

Programming Language: C

Compiler: GCC

Operating System: Linux (Ubuntu)

5. Files Used

main.c

lexer.c

lexer.h

Input file

Output screenshot

6. Working

The program reads the input file, separates it into tokens, compares each token with predefined patterns and keywords, and displays the token type.

7. Output

Displays token types such as Keyword, Identifier, Operator, Constant, String Literal, and Special Symbol.

8. Applications

Compiler design, syntax analysis, source code parsing, and programming language processing.

9. Advantages

Simple, fast, and useful for understanding compiler design concepts.

10. Future Enhancements

Support comments, preprocessor directives, lexical error detection, and symbol table generation.

11. Conclusion

The project successfully identifies lexical tokens from a C source file and demonstrates the lexical analysis phase of a compiler.